



Modbus communication protocol

RESTRICTED DISTRIBUTION

INVERTER IMEON 3.6 / IMEON 9.12

Modbus communication protocol

Notifications index

Index	Date	Changed pages	Description of modification made	Author
A.1	09/12/2015	-	Initial writing	F.L.R.
A.2	02/03/2016	-	Add writing protocols	Y.A.A
A.3	30/01/2017	-	Correct protocols	Y.A.A
A.4	10/02/2017	-	Add protocols	Y.A.A
A.5	07/03/2017	-	Correct protocols	Y.A.A
A.6	03/04/2017	10	Add protocol to send power from Smart Meter	Y.A.A
A.7	11/04/2017	10,11	Add protocol to send power from Smart Meter to every phase	Y.A.A
A.8	19/06/2017	7,11	Add Identification frame to activate Modbus communication with Imeon	Y.A.A
A.9	05/07/2017	8-13	Add list of errors and warnings. Add protocol to read battery consumption and stored power. Add more examples.	Y.A.A
A.10	04/10/2017	-	Rework on the protocol	R.L.F
A.11	05/10/2017	-	Add IMEON specification and example	RLF
A.12	10/11/2017	-	Add function 0x11 to make a write with a return status and 0x10 to make a write without return status, add new read / write function	RLF
A.13	22/12/2017	-	Add new writing function, specify special case when working with lithium battery	RLF
A.14	05/02/2018	22,27	Add settable discharge current range, correction warning code	RLF
A.15	07/03/2018	-	Add read protocols about Smart Meter and energy distribution, add settable max average voltage in inverter	Y.A.A
A.16	26/04/2018	22,23	Add MPPT reading/setting	Y.A.A
A.17	13/07/2018	5,15,23	Add disclaimer for external use, modification of type commande 0x1324, correction example 0x1418. Change of distribution statut (confidential → restricted)	R.L.F
A.18	24/07/2018	5,25	Modification disclaimer, correction value for programmable range	R.L.F
A.19	27/07/2018	5,6,26	Modification disclaimer, correction example 0x1303, 0x1304, 0x1305.	Y.A.A
A.20	21/09/2018	6,21,31	Modification 0x1321, commande write only.	R.L.F
A.21	14/11/2018	5,6,19,30	§1.1-Specification of sending rate. §1.2 –Specification of database refresh period. §2.10 – Add information for 0x1306. §4: Code 1 is deleted (not applicable)	L.K
A.22	08/02/2018	18	§1.3-Add recommendation about cabling. §2.10- Update for register adress 0x1100 (correction of associated example)	L.K
A.23	07/03/2019	28	Correction of 0x1418 command description	RLF
A.24	08/04/2019	5; 11-14	Minial software version is updated (§1), registers for history of production and consumption are removed (§2).	L.K
A.25	06/06/2019	-	Minial software version is updated (§1), update size for register 0x0015 (Main CPU software version), update examples for registers, add register for DRED.	L.K
A.26	13/02/2020	5; 6	Update of §1.2.	L.K
A.27	10/03/2020	9;11-15;18;33;46	Fix currents, pv voltages, AC frequencies, max injection grid power, power factor description units and rounding, highlight write constraint for management status, add detailed overheating in errors list, add newer certificates and fix 0x000C label in 0x1322, add response timeout	T.H
A.28	24/03/2020		Add explanation for meter correction usage; add 0x1421 VPP command	T.H
A.29	04/06/2020	26	Targetted per phase power on inverter grid connector 0x1423	R.L.F

References	Modbus communication protocol – RW – EN – IMEON 3.6 / IMEON 9.12	Index	A.29
------------	--	-------	------

Summary

Summary	3
List of tables	5
List of figures	5
1. Modbus communications protocol for IMEON 3.6 / IMEON 9.12 inverter	6
1.1 Setting parameters	6
1.2 Lithium battery settings	6
1.3 RS485 connections for IMEON 3.6 / IMEON 9.12	7
Table 1. RJ45 pin assignment.	7
1.4 IMEON Modbus protocol message format	7
1.5 Protocol for the read function	8
1.5.1 Query	8
1.5.2 Response	8
1.6 Protocol for the write function	9
1.6.1 Query	9
1.6.2 Response	10
2. Modbus protocol detailed	11
2.1 IDENTIFICATION	11
2.2 GENERAL INFORMATION	11
2.3 AC GRID	11
2.4 AC OUTPUT	13
2.5 INPUT/OUTPUT BATTERY	14
2.6 PV CONNECTION	14
2.7 ERRORS / WARNINGS	15
2.8 SMART METER	15
2.9 AUTO TEST FOR CEI COMPLIANCE	16
2.10 WRITING PROTOCOL	18
3. Information about writable parameters	27
3.1 Programmable range for write command	27
3.2 Grid connection specification	27
3.3 Battery setting specification	28
Figure 2. Battery setting order	28
3.4 Meter correction	29
Figure 3. Meter correction behaviour	29
3.5 Frequency and Voltage results of the Auto test for the CEI compliance	30

4. Error/Warning codes	32
5. DRED mapping.....	35
Do not generate power.	35
1. Disable the feed-in to the grid with register address 0x1306 (bit 13).	35
6. Frame examples	37

RESTRICTED

List of tables

Table 2. Protocol for read function, query format.....	8
Table 3. Protocol for read function, response format	8
Table 4. Protocol for write function, query format.....	9
Table 5. Protocol for write function, answer format	10
Table 6. List of GENERAL INFORMATION and AC GRID addresses.....	12
Table 7. List of AC OUTPUT addresses.	13
Table 8. List of INPUT/OUTPUT BATTERY addresses.	14
Table 9. List of INPUT/OUTPUT PV CONECTION addresses.	14
Table 10 List of Error/Warnings addresses.....	15
Table 11 Smart Meter.....	15
Table 12 List of CEI test addresses.....	17
Table 13. List of WRITING PROTOCOL addresses.	26
Table 14. Programmable range for write command per model.....	27
Table 15. Frequency and Voltage results of the Auto test.....	31
Table 16 Fault/Warning codes	34

List of figures

Figure 1. RJ45 connector pinout	7
Figure 2. Battery setting order.....	28

This protocol works with inverters having a serial number starting by:

For 3.6 : 96131608XXXXXX

For 9.12 : 8809111XXXXXX

To use Modbus protocol, you need an activation key that is sent to the address 0x9000.

If you do not have this activation Key, please contact IMEON.

The use of this protocol affects the overall operation of the inverter and therefore must be carried out by competent professionals.

- The user takes full responsibility for any and all consequences of misuse. IMEON ENERGY can not be held liable for damage of the inverter, batteries or any component of the installation.
- No support will be given for development.

1. Modbus communications protocol for IMEON 3.6 / IMEON 9.12 inverter

This document describes the different addresses used for the communication on Modbus protocol for the inverter IMEON 3.6 and 9.12.

The minimal software version for using this protocol is 1.7.8.2.

1.1 Setting parameters

The setting from the OS. ONE and the setting you change with this protocol both have the same level of priority and will be executed by an order of arrival.

The parameters set by the protocol aren't saved, if you restart the inverter, the parameters will come back to their previous states.

Since your settings are not saved in the inverter, they are not shown in the OS. ONE interface.

When you send a command, wait until the slave response you before sending a new command. The time between the sent of two read commands must be higher than 500ms.

The time between the sent of two write commands must be higher than 1 second.

1.2 Lithium battery settings

If you are using a lithium battery with the inverter, you need to manage its BMS.

Please select battery : "Not used" from the Battery screen drop-down list in OS. ONE interface to prevent conflicts with IMEON's EMS, if you need to adjust any battery setting by yourself. More information is available in section [3.3 Battery setting specification](#).

1.3 RS485 connections for IMEON 3.6 / IMEON 9.12

The IMEON RS485 port is configured to use the following communication setting:

- Baud rate : 9600
- Data bits per Byte : 8
- Parity : None
- Stop bits : 1
- Encoding: big endian(MSB LSB: value 0x1234 is 0x1234)
- CRC: little endian (LSB MSB: value 0x3412 is 0x1234)
- Slave address: 0x01
- Data message format: Hexadecimal
- The instant read values in the database are updated each 10 seconds.
- The set values are updated each 1 minute in database.
- Response timeout is 10 seconds.

The IMEON RS485 connector is wired as below. The length of the cable must be a maximum of 10 to 20m depending on the cable category used (stranded cable of minimum category 6A is recommended). In case a longer cable used, please refer to the standard IEA-485 and the guideline TSB-89.

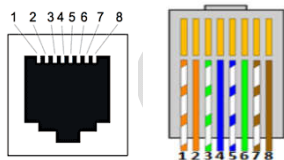


Figure 1. RJ45 connector pinout.

Pin	Function
1	A
2	B
8	GND

Table 1. RJ45 pin assignment.

1.4 IMEON Modbus protocol message format

The Modbus version used is RTU (mode master/slave).

The Modbus protocols define a format for the master query and the slave response, the IMEON is defined as the slave for the communication.

1.5 Protocol for the read function

1.5.1 Query

First Byte

Last Byte

Slave address	Command type	Register address	Function word	Error check (CRC)
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes

Example: query to get IEMON serial number

0x01	0x03	0x0001	0x0004	0x15C9
------	------	--------	--------	--------

Table 2. Protocol for read function, query format

Slave address: the address of the slave being called, 0x01 for IMEON.

Command type: the code of the command being called, for the example 0x03 for reading.

Register address: the address of the register the command is used on, for the example it's 0x0001 for IMEON serial number.

Function words: the number of words required by the function, a word is made of 2 Bytes, for the example it's 0x0004 for IMEON serial number as it returns a value on 8 Bytes.

Error check (CRC): the CRC is calculated based on the value of all the other Bytes from the query, for the example query the CRC is 0xC915 which converts to 0x15C9.

In Modbus, you can read multiple values if the register addresses follow one another by indicating the first register and the number of words to read.

For example, to read the AC grid voltage of all 3 phases, in the query, define the register address as 0x0100 and function words as 0x0003. The response will then contain the value for each phase.

1.5.2 Response

First Byte

Last Byte

Slave address	Command type	Number of Bytes	Response data	Error check (CRC)
1 Byte	1 Byte	1 Bytes	X Bytes	2 Bytes

Example: response answer for the IMEON serial number query

0x01	0x03	0x08	0x00 00 00 00 00 00 00 00	0x95 D7
------	------	------	------------------------------	---------

Table 3. Protocol for read function, response format

Example serial value is set to 0 for illustration.

Slave address: the address of the slave being called, 0x01 for IMEON.

Command type: the code of the command being called, for the example 0x03 for reading.

Number of Bytes: the number of Bytes of data, for example 0x08 for the Serial Number.

Response data: the data return by the function. The number of Bytes returned depends on the function. The returned Bytes are in the format indicated in the description of the function. For the serial number, the format is a “long int” on 8 Bytes which represent the serial number value.

Error check (CRC): the CRC is calculated based on the value of all the other Bytes from the response, for the example response the CRC is 0xD795 which converts to 0x95D7.

1.6 Protocol for the write function

When you send a write command, you should take into account that the inverter can require some time in order to apply the new parameter.

1.6.1 Query

First Byte						Last Byte
Slave address	Command type	Register address	Function words	Data Bytes	Data value	Error check (CRC)
1 Byte	1 Byte	2 Bytes	2 Bytes	1 Byte	X Bytes	2 Bytes

Example: query to set the time to charge the battery by AC grid from 05:00 to 08:30 without returned status.

0x01	0x10	0x1309	0x0002	0x04	0x05 0x00 0x08 0x1E	0x6DF1
------	------	--------	--------	------	------------------------	--------

Example: query to set the time to charge the battery by AC grid from 05:00 to 08:30 with returned status.

0x01	0x11	0x1309	0x0002	0x04	0x05 0x00 0x08 0x1E	0x3C34
------	------	--------	--------	------	------------------------	--------

Table 4. Protocol for write function, query format

Slave address: the address of the slave being called, 0x01 for IMEON.

Command type: the code for the command being called, for the example 0x10 for writing.

Register address: the address of the register the command is used on, for the example 0x1309 to set the time to charge the battery by AC grid.

Function words: the number of words required by the function, rounded up, i.e. if a function requires 7 Bytes the query requires 4 words. For the example, the function requires 4 Bytes therefore 2 words: 0x002.

Data Bytes: the number of Bytes of data. For the example, we use 4 Bytes of data for the function to set the time to charge the battery by AC grid.

Data value: the value for the Byte we want to write. The format of the data is indicated in the function's description. If multiple values are required for the function, the data needs to be split. For the example, the function is written on 4 Bytes in the following format:

[Hour start] [Minute start] [Hour end] [Minute end]

The example concerns setting from 05:00 to 08:30,

The data value in hexadecimal is therefore "0x05 0x00 0x08 0x1E".

Error check (CRC): the CRC is calculated based on the value of all the other Bytes from the query, for the example response the CRC is 0x6DF1 which converts to 0xF16D.

1.6.2 Response

First Byte

Last Byte

Slave address	Command type	Register address	Function words	Error check (CRC)
1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes

Example: response to set the time to charge the battery by AC grid from 05:00 to 08:30 without status.

0x01	0x10	0x1309	0x0002	0xA88E
------	------	--------	--------	--------

Slave address	Command type	Register address	Function words	Status	Error check (CRC)
1 Byte	1 Byte	2 Bytes	2 Bytes	1 Bytes	2 Bytes

Example: response to set the time to charge the battery by AC grid from 05:00 to 08:30 with status.

0x01	0x11	0x1309	0x0002	0x80	0x8E78
------	------	--------	--------	------	--------

Table 5. Protocol for write function, answer format

Slave address: the address of the slave being called, 0x01 for IMEON.

Command type: the code for the command being called, for the example 0x10 for writing.

Register address: the address of the register the command is used on, for the example 0x1309 to set the time to charge the battery by AC grid.

Function words: the number of words required by the function. For the example, the function needs 4 Bytes therefore 2 words: 0x002.

Status: the status of the function used, there are 3 possible outcomes:

- 0xFF: indicates an error in the query, either the query is invalid or the data doesn't fit the function.

- 0x00: the query was accepted but the inverter refused the data, for example due to a conflict with another setting.

- 0x80: the query was accepted and the inverter accepted the setting.

In the example, 0x80 mean that the setting was accepted.

Error check (CRC): the CRC is calculated based on the value of all the other Bytes from the query, for the example response the CRC is 0x8FCF which converts to 0xCF8F.

2. Modbus protocol detailed

The following tables show the different function codes used for Modbus protocol.

- **Master** : Device
- **Slave** : IMEON

The addresses with* (**asterisk**) can be used only for IMEON 9.12 inverter. When used for IMEON 3.6 inverter, these addresses will return 0xFF.

The addresses with ** (**double asterisk**) can be used only for IMEON 3.6 inverter. When used for IMEON 9.12 inverter, these addresses will return 0xFF.

Register address	Function word	Function	Response/Query Format	Command Type
2.1 IDENTIFICATION				
0x9000	4	Identification frame to activate the Modbus communication	8 Bytes key provided by IMEON 0xXX 0xXX 0xXX 0xXX 0xXX 0xXX 0xXX 0xXX Ex: Send : 01 10 90 00 00 04 08 01 00 00 00 00 00 00 00 B6 E7 Receive : 01 10 90 00 00 04 1C EE	W
2.2 GENERAL INFORMATION				
0x0001	4	S/N IMEON	"long int" encoded on 8 Bytes 0xXX XX XX XX XX XX XX XX Ex: 88888832053200 Send : 01 03 00 01 00 04 15 C9 Receive : 01 03 08 00 00 50 D8 0B 42 4F D0 2E 08	R
0x0015	3	Main CPU software version	ASCII value, each char is encoded on a Byte. Ex: 1.7.8.0 Send : 01 03 00 15 00 07 15 CC Receive : 01 03 0E 30 31 2E 30 30 37 2E 30 30 38 2E 30 30 30 73 63 Value in Ascii : '01.007.008.000'	R
2.3 AC GRID				
0x0100	1	AC Grid voltage L1	"short" encoded on 2 Bytes, the value has a resolution of 0.1V Ex: 231.1V >> 0x09 0x07(2311)	R
0x0101*	1	AC Grid voltage L2	"short" encoded on 2 Bytes, the value has a resolution of 0.1V	R

Register address	Function word	Function	Response/Query Format	Command Type
0x0102*	1	AC Grid voltage L3	"short" encoded on 2 Bytes, the value has a resolution of 0.1V	R
0x0103	2	AC Grid power L1 (From Smart Meter plus Grid connector of inverter plus consumed grid to charge batteries)	Signed "int" encoded on 4 Bytes, the value has a resolution of 1W Ex : 955W 0x00 0x00 0x03 0xBB (955) Ex : -955W 0xFF 0xFF 0xFC 0x45 (-955)	R
0x0105*	2	AC Grid power L2 (From Smart Meter plus Grid connector of inverter plus consumed grid to charge batteries)	Signed "int" encoded on 4 Bytes, the value has a resolution of 1W Ex: 77W 0x00 0x000x00 0x4D (77) Ex: -77W 0xFF 0xFF 0xFF 0xB3 (-77)	R
0x0107*	2	AC Grid power L3 (From Smart Meter plus Grid connector of inverter plus consumed grid to charge batteries)	Signed "int" encoded on 4 Bytes, the value has a resolution of 1W Ex: 77W 0x00 0x000x00 0x4D (77) Ex: -77W 0xFF 0xFF 0xFF 0xB3 (-77)	R
0x0109*	2	AC Grid power (total) (From Smart Meter plus Grid connector of inverter plus consumed grid to charge batteries)	Signed "int" encoded on 4 Bytes, the value has a resolution of 1W Ex: 77W 0x00 0x000x00 0x4D (77) Ex: -77W 0xFF 0xFF 0xFF 0xB3 (-77)	R
0x0117	1	AC Grid frequency	"short" encoded on 2 Bytes, the value has a resolution of 0.1 Hz Ex: 1Hz 0x00 0x64 (100)	R
0x0118	1	AC Grid current L1	"short" encoded on 2 Bytes, the value has a resolution of 0.1A Ex: 4.5A 0x00 0x2D(45)	R
0x0119*	1	AC Grid current L2	"short" encoded on 2 Bytes, the value has a resolution of 0.1A	R
0x011A*	1	AC Grid current L3	"short" encoded on 2 Bytes, the value has a resolution of 0.1A	R
0x011B*	1	AC Grid current (total)	"short" encoded on 2 Bytes, the value has a resolution of 0.1A Ex : 1A 0x00 0x0A (10)	R
0x011C	2	Maximum grid injection power	"int" encoded on 4 Bytes, the value has a resolution of 1W Ex: 77W 0x00 0x00 0x00 0x4D (77)	R
0x011E	1	AC Grid power factor	"short" encoded on 2 Bytes, the value has a resolution of 0.01, value goes from 0.9 to 1 and -0.9 to -0.99, it translates to a value from 90 to 100 for positive and 190 to 199 for negative. Ex: 0.91 0x00 0x5B (91) Ex: -0.92 0x00 0xC0 (192)	R

Table 6. List of GENERAL INFORMATION and AC GRID addresses.

Register address	Function word	Function	Response/Query Format	Command Type
2.4 AC OUTPUT				
0x0200	1	AC output voltage L1	"short" encoded on 2 Bytes, the value has a resolution of 0.1V Ex: 234.6V 0x09 0x2A (2346)	R
0x0201*	1	AC output voltage L2	"short" encoded on 2 Bytes, the value has a resolution of 0.1V	R
0x0202*	1	AC output voltage L3	"short" encoded on 2 Bytes, the value has a resolution of 0.1V	R
0x0203	2	AC output power L1 (From AC output connector of inverter)	Signed "int" encoded on 4 Bytes, the value has a resolution of 1W Ex: 967W 3. 0x00 0x00 0x03 0xC7 (967) 4. Ex: -967W 0xFF 0xFF 0xFC 0x39 (-967)	R
0x0205*	2	AC output power L2 (From AC output connector of inverter)	Signed "int" encoded on 4 Bytes, the value has a resolution of 1W	R
0x0207*	2	AC output power L3 (From AC output connector of inverter)	Signed "int" encoded on 4 Bytes, the value has a resolution of 1W	R
0x0209*	2	AC output power(total) (From AC output connector of inverter)	Signed "int" encoded on 4 Bytes, the value has a resolution of 1W	R
0x0211	1	AC output frequency	"short" encoded on 2 Bytes, the value has a resolution of 0.1Hz	R
0x0212	1	AC output current L1	"short" encoded on 2 Bytes, the value has a resolution of 0.1A Ex: 4.5A 0x00 0x2D(45)	R
0x0213*	1	AC output current L2	"short" encoded on 2 Bytes, the value has a resolution of 0.1A	R
0x0214*	1	AC output current L3	"short" encoded on 2 Bytes, the value has a resolution of 0.1A	R
0x0215*	1	AC output current (total)	"short" encoded on 2 Bytes, the value has a resolution of 0.1A	R

Table 7. List of AC OUTPUT addresses.

Register address	Function word	Function	Response/Query Format	Command Type
4.1 INPUT/OUTPUT BATTERY				
0x0300	1	Battery voltage	"short" encoded on 2 Bytes, the value has a resolution of 0.1V Ex: 50.8V 0x01 0xFC (508)	R
0x0301	1	Battery charge current	"short" encoded on 2 Bytes, the value has a resolution of 0.1A Ex: 2.5A 0x00 0x19 (25)	R
0x0302	1	Battery discharge current	"short" encoded on 2 Bytes, the value has a resolution of 0.1A Ex: 95A 0x03 0xB6 (950)	R
0x0303	1	Battery SOC (state of charge)	"short" encoded on 2 Bytes, the value has a resolution of 1% Ex: 90% 0x00 0x5A(90)	R

Table 8. List of INPUT/OUTPUT BATTERY addresses.

Register address	Function word	Function	Response/Query Format	Command Type
4.2 PV CONNECTION				
0x0400	2	PV1 power	"int" encoded on 4 Bytes, the value has a resolution of 1W Ex: 4250W 0x00 0x00 0x10 0x9A (4250)	R
0x0402*	2	PV2 power	"int" encoded on 4 Bytes, the value has a resolution of 1W	R
0x040A	1	PV1 voltage	"short" encoded on 2 Bytes, the value has a resolution of 0.1V Ex: 373.4V 0x0E 0x96 (3734)	R
0x040B*	1	PV2 voltage	"short" encoded on 2 Bytes, the value has a resolution of 0.1V	R
0x040C*	2	Total PV power	"int" encoded on 4 Bytes, the value has a resolution of 1W Ex: 4250W 0x00 0x00 0x10 0x9A (4250)	R
0x040E*	1	PV1 current	"short" encoded on 2 Bytes, the value has a resolution of 0.1A	R
0x040F*	1	PV2 current	"short" encoded on 2 Bytes, the value has a resolution of 0.1A Ex: 6.2A 0x00 0x3E (62)	R

Table 9. List of INPUT/OUTPUT PV CONNECTION addresses.

Register address	Function word	Function	Response/Query Format	Command Type
4.3 ERRORS / WARNINGS				
0x0503	2	ErrorType	The last four recorded errors. The format is "int" encoded on 1 Byte for every error. To see the list of all errors, please see the table "Error/Warning codes" in section 4. Ex : 0x42 0x00 0x00 0x00 [42]: Solar input overvoltage. [00]: Not a fault	R
0x0505	4	Warning Type	The last eight recorded warnings. The format is "int" encoded on 1 Byte for every warning. To see the list of all warnings, please see the table "Error/Warning codes" in section 4. Ex : 0x02 0x00 0x00 0x00 0x00 0x00 0x00 0x00 [02]: Islanding detected. [00]: Not a fault	R

Table 10 List of Error/Warnings addresses.

Register address	Function word	Function	Response/Query Format	Command Type
4.4 SMART METER				
0x0600	2	Smart Meter power	"int" encoded on 4 Bytes, the value has a resolution of 1W Ex: 4250W 0x00 0x00 0x10 0x9A (4250) Ex: -4250W 0xFF 0xFF 0xEF 0x66 (-4250)	R
0x0602	1	Smart Meter status	"short" encoded on 2 Bytes Two possible values: 0 : Not connected/Not communication with smart meter 1 : Communication with smart meter	R

Table 11 Smart Meter

Register address	Function word	Function	Response/Query Format	Command Type
4.5 AUTO TEST FOR CEI COMPLIANCE work only when CEI compliance is set				
0x1324	1	Start auto test for CEI compliance	“short” encoded on 2 Bytes. To start the Auto test send value 0x0000 Ex: Send : 01 10 13 24 00 01 02 00 00 82 75 Receive : 01 10 13 24 01 36 05	W
0x1325	20	Get results of frequency of CEI autotest	This value can be obtained 30 seconds after the start time of the auto test To see the description of the results given by the inverter, please see the table “Results of Frequency test” on 3.4 section Ex : [12 8E] [00 64] [01 F3] [00 32] [00 64] [02 03] [00 64] [13 87] [13 54] [00 64] [13 56] [00 64] [13 81] [00 32] [00 64] [13 BA] [00 64] [13 87] [13 81] [00 64] TestStep: 81<S2 Frequency Test limitFrequency: 47.5Hz; limitTime: 100.0ms; realFrequency: 49.99Hz; tripFrequency: 50.0Hz; tripTime: 100.0ms TestStep: 81>s2 Frequency Test limitFrequency: 51.5Hz; limitTime: 100.0ms; realFrequency: 49.99Hz; tripFrequency: 49.98Hz; tripTime: 100.0ms TestStep: 81<S1 Frequency Test limitFrequency: 49.5Hz; limitTime: 100.0ms; realFrequency: 49.99Hz; tripFrequency: 50.0Hz; tripTime: 100.0ms TestStep: 81>s1 Frequency Test limitFrequency: 50.5Hz; limitTime: 100.0ms; realFrequency: 49.99Hz; tripFrequency: 49.98Hz; tripTime: 0.0ms	R

Register address	Function word	Function	Response/Query Format	Command Type
0x1345	20	Get results of the voltage in phase L1 of CEI autotest	This value can be obtained 30 seconds after the start time of the auto test To see the description of the results given by the inverter, please see the table “Results of Frequency test” on section 3.4 Ex : [07 9E] [01 90] [08 8F] [08 90] [01 90] [0A 55] [00 C8] [08 8F] [08 8F] [00 C8] [00 5C] [00 C8] [08 8F] [08 90] [00 C8] [00 FD] [0B B8] [08 8F] [08 8E] [0B B8] TestStep: 81<S2 Volt Test limitVoltage: 195.0V; limitTime: 400.0ms; realVoltage: 219.1V; tripVoltage: 219.2V; tripTime: 400.0ms TestStep: 81>S2 Volt Test limitVoltage: 264.5V; limitTime: 200.0ms; realVoltage: 219.1V; tripVoltage: 219.0V; tripTime: 200.0ms TestStep: 81<S1 Volt Test limitVoltage: 9.2V; limitTime: 200.0ms; realVoltage: 219.1V; tripVoltage: 219.2V; tripTime: 200.0ms TestStep: 81>S1 Volt Test limitVoltage: 253.0V; limitTime: 3000.0ms; realVoltage: 219.1V; tripVoltage: 219.0V; tripTime: 3000.0ms	R
0x1365	20	Get results of the voltage in phase L2 of CEI autotest	This value can be obtained 30 seconds after the start time of the auto test To see the description of the results given by the inverter, please see the table “Results of Voltage test” on 3.4 section	R
0x1385	20	Get results of the voltage in phase L3 of CEI autotest	This value can be obtained 30 seconds after the start time of the auto test To see the description of the results given by the inverter, please see the table “Results of Frequency test” on 3.4 section	R

Table 12 List of CEI test addresses.

Register address	Function word	Data Byte	Function	Response/Query Format	Command Type
4.6 WRITING PROTOCOL					
0x1000	1	2	LCD sleep time in seconds 0 (always on) / 30 / 60 / 300 / 600	“short” encoded on 2 Bytes Ex: 60s 0x00 0x3C (60)	R/W
0x1001	4	7	Date and time on 24H format	Return 8 Bytes, last Byte is 0 and is unused. [Year][Month][Day]Hour[Minute][Second] Ex: 4/10/2017 10:54:00 0x07E1 0x0A 0x04 0x0A 0x36 0x00 ([2017] [10] [04] [10] [54] [00])	R/W
0x1100	2	4	Maximum grid injection power	“int” encoded on 4 Bytes, the value has a resolution of 1W Ex: 9000W 0x00 0x00 0x23 0x28 (9000)	R/W
0x1300	2	4	Battery voltage and Battery SOC ⁽¹⁾	Two “short” encoded on 4 Bytes, the resolution for the voltage is 0.1V and for the SOC is 1%. [voltage][SOC] Ex : 49.8V 99% 0x01 0x F2 0x00 0x63 ([498][99])	R/W
0x1303	1	2	Max. battery charging current by PV	“short” encoded on 2 Bytes, the value has a resolution of 0.1A Ex: 25 A 0x00 0xFA (250)	R/W
0x1304	1	2	Max. battery charging current by AC Grid	“short” encoded on 2 Bytes, the value has a resolution of 0.1A Ex: 10A 0x00 0x64 (100)	R/W
0x1305	1	2	Max. battery discharge current limit	“short” encoded on 2 Bytes, the value has a resolution of 0.1A Ex: 15 A 0x00 0x96 (150)	R/W

Register address	Function word	Data Byte	Function	Response/Query Format	Command Type																																				
0x1306	1	2	IMEON power management setting	For read function, the state for each item is indicated as a flag for the given bit, 0 for disabled and 1 for enabled.	R/W																																				
				For the write function, you can <u>set only one value per request</u> . To enable or disable one of the items send one of the value given below.																																					
				<table><tr><td></td><td>position</td><td>Enable</td><td>Disable</td></tr><tr><td>charge battery with PV</td><td>bit 15</td><td>0x8000</td><td>0x7FFF</td></tr><tr><td>charge battery with AC grid</td><td>bit 14</td><td>0x4000</td><td>0xBFFF</td></tr><tr><td>feed in to AC grid</td><td>bit 13</td><td>0x2000</td><td>0xDFFF</td></tr><tr><td>discharge battery to loads when PV present</td><td>bit 12</td><td>0x1000</td><td>0xEFFF</td></tr><tr><td>discharge battery to loads when PV absent</td><td>bit 11</td><td>0x0800</td><td>0xF7FF</td></tr><tr><td>discharge battery to AC grid when PV present</td><td>bit 10</td><td>0x0400</td><td>0xFBFF</td></tr><tr><td>discharge battery to AC grid when PV absent</td><td>bit 09</td><td>0x0200</td><td>0xFDFF</td></tr><tr><td>AC output*</td><td>bit 08</td><td>0x0100</td><td>0xFEFF</td></tr></table>			position	Enable	Disable	charge battery with PV	bit 15	0x8000	0x7FFF	charge battery with AC grid	bit 14	0x4000	0xBFFF	feed in to AC grid	bit 13	0x2000	0xDFFF	discharge battery to loads when PV present	bit 12	0x1000	0xEFFF	discharge battery to loads when PV absent	bit 11	0x0800	0xF7FF	discharge battery to AC grid when PV present	bit 10	0x0400	0xFBFF	discharge battery to AC grid when PV absent	bit 09	0x0200	0xFDFF	AC output*	bit 08	0x0100	0xFEFF
						position	Enable	Disable																																	
				charge battery with PV		bit 15	0x8000	0x7FFF																																	
				charge battery with AC grid		bit 14	0x4000	0xBFFF																																	
				feed in to AC grid		bit 13	0x2000	0xDFFF																																	
				discharge battery to loads when PV present		bit 12	0x1000	0xEFFF																																	
				discharge battery to loads when PV absent		bit 11	0x0800	0xF7FF																																	
				discharge battery to AC grid when PV present		bit 10	0x0400	0xFBFF																																	
				discharge battery to AC grid when PV absent		bit 09	0x0200	0xFDFF																																	
				AC output*		bit 08	0x0100	0xFEFF																																	
				* Ac Output enable can last up to 20 minutes, this time depending on grid norm applied.																																					
				Ex: Send : 01 03 13 06 00 01 60 8F Receive : 01 03 02 F9 00 FA 14 Value F9 to binary [1111 1001 0000 0000] Means :																																					
				<table><tr><td></td><td>Value</td></tr><tr><td>Charge battery with PV</td><td>1</td></tr><tr><td>Charge battery with AC grid</td><td>1</td></tr><tr><td>Feed in to AC grid</td><td>1</td></tr><tr><td>Discharge battery to loads when PV present</td><td>1</td></tr><tr><td>Discharge battery to loads when PV absent</td><td>1</td></tr><tr><td>Discharge battery to AC grid when PV present</td><td>0</td></tr><tr><td>Discharge battery to AC grid when PV absent</td><td>0</td></tr><tr><td>AC output</td><td>1</td></tr></table>			Value	Charge battery with PV	1	Charge battery with AC grid	1	Feed in to AC grid	1	Discharge battery to loads when PV present	1	Discharge battery to loads when PV absent	1	Discharge battery to AC grid when PV present	0	Discharge battery to AC grid when PV absent	0	AC output	1																		
						Value																																			
				Charge battery with PV		1																																			
Charge battery with AC grid	1																																								
Feed in to AC grid	1																																								
Discharge battery to loads when PV present	1																																								
Discharge battery to loads when PV absent	1																																								
Discharge battery to AC grid when PV present	0																																								
Discharge battery to AC grid when PV absent	0																																								
AC output	1																																								

Register address	Function word	Data Byte	Function	Response/Query Format	Command Type
0x1307	1	2	Battery bulk charge voltage	“short” encoded on 2 Bytes, the value has a resolution of 0.1V Ex: 57V 0x02 0x3A (570)	R/W
0x1308	1	2	Battery floating charge voltage	“short” encoded on 2 Bytes, the value has a resolution of 0.1V Ex: 55.4V 0x02 0x2A (554)	R/W
0x1309	2	4	Time to charge battery by AC Grid (24H format is used)	Four “char” encoded on 4 Bytes, the format is as below: [hour start] [minute start] [hour end] [minute end] Ex: from 05:00 to 08:30 0x05 0x00 0x08 0x1E ([05] [00] [08] [30])	R/W
0x130B	1	2	Battery cut-off voltage when grid is absent	“short” encoded on 2 Bytes, the value has a resolution of 0.1V Ex: 47V 0x01 0xD6 (470)	R/W
0x130C	1	2	Battery re-discharge voltage when grid is absent	“short” encoded on 2 Bytes, the value has a resolution of 0.1V	R/W
0x130D	1	2	Battery cut-off voltage when grid is present	“short” encoded on 2 Bytes, the value has a resolution of 0.1V	R/W
0x130E	1	2	Battery re-discharge voltage when grid is present	“short” encoded on 2 Bytes, the value has a resolution of 0.1V	R/W
0x130F	1	2	Battery cut-off SOC when grid is unavailable	“short” encoded on 2 Bytes, the value has a resolution of 1%. Ex: 35% 0x00 0x23 (35)	R/W
0x1310	1	2	Battery cut-off SOC when grid is available	“short” encoded on 2 Bytes, the value has a resolution of 1%.	R/W
0x1311	1	2	Feed-in power factor	“short” encoded on 2 Bytes, the value has a resolution of 0.01. Value goes from 0.9 to 1 and -0.9 to -0.99, it translate to a value from 90 to 100 for positive and 190 to 199 for negative. Ex: 0.99 0x00 0x63 (99)	R/W

Register address	Function word	Data Byte	Function	Response/Query Format	Command Type
0x1312	1	2	Min. AC grid connected frequency	“short” encoded on 2 Bytes, the value has a resolution of 0.1Hz. Ex: 47.6Hz 0x01 0xDC (476)	R/W
0x1313	1	2	Max. AC grid connected frequency	“short” encoded on 2 Bytes, the value has a resolution of 0.1Hz.	R/W
0x1314	1	2	Min. AC grid connected voltage	“short” encoded on 2 Bytes, the value has a resolution of 0.1V Ex: 189.0V 0x07 0x62 (1890)	R/W
0x1315	1	2	Max. AC grid connected voltage	“short” encoded on 2 Bytes, the value has a resolution of 0.1V	R/W
0x1316	3	6	Control power with Smart Meter ⁽¹⁾	“long int” encoded on 6 Bytes. The resolution is 1W. To increase the injection value we send the value of power at which we add 100 000. For reducing the injection, we send the power we want to reduce. Ex: reduce injection by 50, we send 50 0x00 0x00 0x00 0x00 0x00 0x32 (50) Increase injection by 125, we send 100125 0x00 0x00 0x00 0x01 0x87 0x1D (10025)	W
0x1319	3	6	Control power with Smart Meter for phase L1 ⁽¹⁾	“long int” encoded on 6 Bytes. The resolution is 1W. To increase the injection value we add a ‘1’ in front of the value written on 5 digit, which come to add 100 000 to the value. Ex: Increase injection by 125, we send 100125 0x00 0x00 0x00 0x01 0x87 0x1D (10025) For reducing the injection, we add a ‘0’ in front of the value we want to reduce. It is equivalent to send the value. Ex: reduce injection by 50, we send 50 0x00 0x00 0x00 0x00 0x00 0x32 (50)	W
0x131C	3	6	Control power with Smart Meter for phase L2 ⁽¹⁾	“long int” encoded on 6 Bytes. The resolution is 1W. Same format than for L1	W

Register address	Function word	Data Byte	Function	Response/Query Format	Command Type																								
0x131F	3	6	Control power with Smart Meter for phase L3 ⁽¹⁾	“longint” encoded on 6 Bytes. The resolution is 1W. Same format than for L1	W																								
0x1321	1	2	Control the relay of connector “Grid Input”	This option is available with AS4777 compliance selected only “short” encoded on 2 Bytes. For the write function, you can set only one value per request. To control the relay, send one of the values given below. <table><tr><th>Value</th><th>Description</th></tr><tr><td>0x0000</td><td>close relay (Normal operation for inverter)</td></tr><tr><td>0x0001</td><td>open relay (Put in standby the inverter)</td></tr></table>	Value	Description	0x0000	close relay (Normal operation for inverter)	0x0001	open relay (Put in standby the inverter)	W																		
Value	Description																												
0x0000	close relay (Normal operation for inverter)																												
0x0001	open relay (Put in standby the inverter)																												
0x1322	1	2	Set compliance on Inverter	This option is only available with inverter firmware version 4.3.1 or over “short” encoded on 2 Bytes. For the write function, you can set only one value per request. To control the relay, send one of the values given below. Ex: Set compliance AS4777. 0x00 0x05 Setting this value takes around 10s due to necessary hardware switching time. <table><tr><th>Value</th><th>Description</th></tr><tr><td>0x0001</td><td>VDE0126</td></tr><tr><td>0x0003</td><td>VDE4105</td></tr><tr><td>0x0005</td><td>AS4777</td></tr><tr><td>0x0007</td><td>G83</td></tr><tr><td>0x0009</td><td>RD1699</td></tr><tr><td>0x000B</td><td>EN 50438</td></tr><tr><td>0x000C</td><td>EN 50438 National deviation Czech Republic</td></tr><tr><td>0x000D</td><td>EN 50438 National deviation Ireland</td></tr><tr><td>0x000E</td><td>NRS 097-2-1</td></tr><tr><td>0x000F</td><td>EN 62109-1-2</td></tr><tr><td>0x0010</td><td>CEI</td></tr></table>	Value	Description	0x0001	VDE0126	0x0003	VDE4105	0x0005	AS4777	0x0007	G83	0x0009	RD1699	0x000B	EN 50438	0x000C	EN 50438 National deviation Czech Republic	0x000D	EN 50438 National deviation Ireland	0x000E	NRS 097-2-1	0x000F	EN 62109-1-2	0x0010	CEI	R/W
Value	Description																												
0x0001	VDE0126																												
0x0003	VDE4105																												
0x0005	AS4777																												
0x0007	G83																												
0x0009	RD1699																												
0x000B	EN 50438																												
0x000C	EN 50438 National deviation Czech Republic																												
0x000D	EN 50438 National deviation Ireland																												
0x000E	NRS 097-2-1																												
0x000F	EN 62109-1-2																												
0x0010	CEI																												

Register address	Function word	Data Byte	Function	Response/Query Format		Command Type										
				0x0011	EN 50438 National deviation Cyprus											
				0x0012	AS4777 National deviation New Zealand											
				0x0013	EN 50438 National deviation Sweden											
				0x0015	VDE0126 National derivation France 2019											
				0x0016	EN 50160 National deviation Austria											
				0x0017	Chile national norm											
0x1323	1	2	Running mode on inverter	“short” encoded on 2 Bytes. For the write function, you can set only one value per request. To set a running mode, send one of the values given below. Ex: Set working mode in smart grid. 0x00 0x01 Setting this value takes around 10s due to necessary hardware switching time. Use the function 0x1306 to change the energy management that is faster as it changes less setting. <table><tr><th>Value</th><th>Description</th></tr><tr><td>0x0001</td><td>Smart Grid</td></tr><tr><td>0x0002</td><td>Back Up</td></tr><tr><td>0x0003</td><td>On Grid</td></tr><tr><td>0x0004</td><td>Off Grid</td></tr></table>		Value	Description	0x0001	Smart Grid	0x0002	Back Up	0x0003	On Grid	0x0004	Off Grid	R/W
Value	Description															
0x0001	Smart Grid															
0x0002	Back Up															
0x0003	On Grid															
0x0004	Off Grid															
0x1405	4	8	Battery cut off and redischage voltage settings	4 “short” encoded each on 2 Bytes, with a value resolution of 0.1V in following order [cut off without grid][redischage without grid][cut off with grid][redischage with grid] Ex: To configure the battery cut off and redischage voltage : Cut off voltage without the grid: 42V convert to 420[0x01A4] Redischage voltage without the grid: 51V convert to 510[0x01F4] Cut off voltage with the grid: 45V convert to 450[0x01C2] Redischage voltage with the grid: 52V convert to 520[0x0208] Message data: 0x01 0xA4 0x01 0xF4 0x01 0xC2 0x02 0x08		R/W										
0x1409	4	8	Battery cut off and redischage SOC settings	4 “short” encoded each on 2 Bytes, with a value resolution 1% in following order [cut off without grid][redischage without grid][cut off with grid][redischage with grid] Ex: To configure the battery cut off and redischage SOC :		R/W										

Register address	Function word	Data Byte	Function	Response/Query Format	Command Type
				Cut off SOC without the grid: 10% convert to 10[0x000A] Redischarge SOC without the grid: 50% convert to 50[0x0032] Cut off SOC with the grid: 20% convert to 20[0x0014] Redischarge SOC with the grid: 85% convert to 85[0x0055] Message data: 0x00 0x0A 0x00 0x32 0x00 0x14 0x00 0x55	
0x140D	5	10	Control injection power with a meter for all phase ⁽¹⁾	This command starts with setting the first Byte at 0, then 3 “Bytes” for the injection sign and 3 “short” encoded on 2 Bytes with a resolution of 1W in following order: [0x00][Phase 1 sign][Phase 1 power] [Phase 2 sign][Phase 2 Power][Phase 3 sign][Phase 3 Power] The sign value is 1 for the “+” and 0 for the “-”. When a “+” is set, the inverter will increase the injected power. When a “-” is set, the inverter will reduce the injected power. Set a power of 0 to keep the injection at the current level. Ex: To increase the injection of +350W on phase 1: [0x01][0x015E] reduce the injection of -1600W on phase 2: [0x00][0x0640] keep the injection on phase 3: [0x00][0x0000] Message Data: 0x00 0x01 0x01 0x5E 0x00 0x06 0x40 0x00 0x00 0x00	W

Register address	Function word	Data Byte	Function	Response/Query Format	Command Type
0x1417	1	2	Restart Operating System	A "short" value encoded on 2 Bytes. This command will restart the operating system of the embedded board. Send a value of 1 to restart the operating system. Ex: To restart the operating system send 1. Message Data: 0x01	W
0x1418	1	2	Max average voltage	"short" encoded on 2 Bytes, the value has a resolution of 0.1V. This value must be set between the range choose for the AC grid connected voltage (0x1314 and 0x1315) Ex: 270.0V 0x0A 0x8C (2700)	R/W
0x1419	2	4	MPPT range voltage	"short" encoded on 4 Bytes, the value has a resolution of 1V. [MPPT voltage low] [MPPT voltage high] This value must be set between the ranges listed on table 14. Ex : 413V and 750V 0x01 0x9D (413) 0x02 0xEE (750)	R/W
0x1421	2	4	Targetted power on inverter grid connector	"short" encoded on 4 Bytes, the value has a resolution of 1W. The value is hold during 1 minute after reception, we recommand resending the command at least every 30s. Sign is Byte, 0x01 for '+' to trigger an injection and 0x00 for '-' to trigger a charge [0x00][sign] [value] Ex: For idle(0W) 0x 00 0x00 0x00 0x00 For injection (+496W) 0x00 0x01 0x01 0xF0 For consumption (-496W) 0x00 0x00 0x01 0xF0	W

Register address	Function word	Data Byte	Function	Response/Query Format	Command Type
0x1423	5	10	Targetted per phase power on inverter grid connector	“short” encoded on 4 Bytes, the value has a resolution of 1W. The value is hold during 1 minute after reception, we recommand resending the command at least every 30s. Sign is Byte, 0x01 for ‘+’ to trigger an injection and 0x00 for ‘-’ to trigger a charge The charge battery will be balanced equally on each phase, we recommand using 0x1421 for consumption. [0x00][sign L1] [value L1]][sign L2] [value L2]][sign L3] [value L3] Ex: Idle mode (0W) 0x01 0x10 0x1423 0x0005 0x0A 0x00 0x00 0x0000 0x00 0x0000 0x00 0x0000 0x37 0x14 Consumption mode (795w) 0x00 0x00 0x0150 0x00 0x0032 0x00 0x0199 Injection mode (I1: +336, I2:+50, I3:+409) 0x00 0x01 0x0150 0x01 0x0032 0x01 0x0199	W

Table 13. List of WRITING PROTOCOL addresses.

(1) This protocol must be send at least every 7 seconds and no more than every 2 seconds.
Some writing commands take time; please take it into account while using the protocol.

5. Information about writable parameters

5.1 Programmable range for write command

	Register address	3.6		9.12	
		min	max	min	max
Max. grid injection power (W)	0x1100	0	3000	0	9000
Max. battery charging current by PV(A)	0x1303	1	60	1	160
Max. battery charging current by AC grid (A)	0x1304	1	60	1	160
Max. battery discharge current (A)	0x1305	11	80	11	200
Battery bulk charge voltage (V)	0x1307	50	62	48	62
Battery floating charge voltage (V)	0x1308	48	62	48	62
Battery cut-off voltage when grid is absent(V)	0x130B	40	51	40	51
Battery re-discharge voltage when grid is absent (V)	0x130C	40	62	40	56
Battery cut-off voltage when grid is present (V)	0x130D	42	48	42	51
Battery re-discharge voltage when grid is present (V)	0x130E	42	62	42	56
Battery cut-off SOC when grid is unavailable (%)	0x130F	5	100	5	100
Battery cut-off SOC when grid is available (%)	0x1310	5	100	5	100
Min. AC grid connected frequency (Hz)	0x1312	45	55	45	55
Max. AC grid connected frequency (Hz)	0x1313	45	55	45	55
Min. AC grid connected voltage (V)	0x1314	180	276	180	276
Max. AC grid connected voltage (V)	0x1315	180	276	180	276
Max average voltage	0x1418	235	275	235	275
MPPT voltage low	0x1419	120	140	350	500
MPPT voltage high	0x1419	400	450	700	750

Table 14. Programmable range for write command per model

5.2 Grid connection specification

The register from 0x1312 to 0x1315 define the AC grid frequency / voltage range within which the inverter stays connected to the AC grid. Those ranges have a minimal value to keep between the programmed min and max values.

For the operating frequency range (reg. 0x1312 and 0x1313) you have to keep at least 1Hz between the min and max.

For the operating voltage range (reg. 0x1314 and 0x1315) you have to keep at least 5V between the min and max.

5.3 Battery setting specification

We can represent the battery settings as bellow:

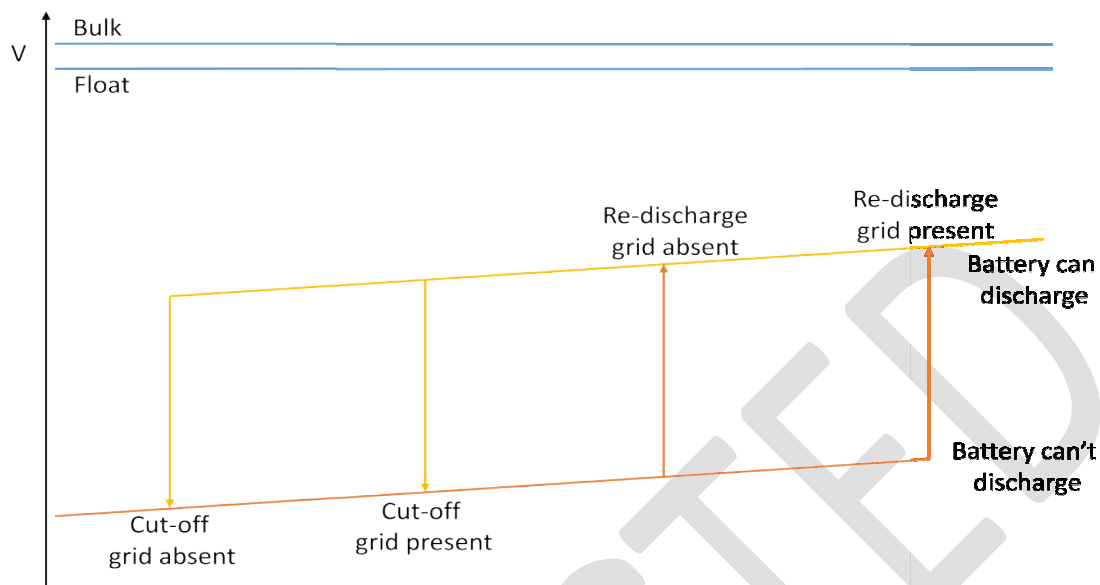


Figure 2. Battery setting order

The cut-off voltage defines a voltage at which IMEON will stop discharging the battery. The re-discharge voltage defines a voltage at which IMEON will discharge the battery again.

If the SOC (expressed in %) and cut-off voltage (expressed in V) differ, the inverter stops discharging the battery as soon as any of the two values is reached. The SOC value is taken into account only if the 0x1300 is used.

The voltage settings for the battery must follow the rules below:

Cut-off < Re-discharge ≤ Float ≤ Bulk.

Cut-off grid absent ≤ Cut-off grid present.

Re-discharge grid absent ≤ Re-discharge grid present.

Re-discharge grid absent can be lower than cut-off grid present.

When changing a value in the battery settings, one has to cross-check the value against the rules above, or risks the command to be rejected.

When increasing any value, it's best to change the battery settings in the following order:

Bulk, Float, Re-discharge grid present, Re-discharge grid absent, Cut-off grid present, Cut-off grid absent.

When decreasing any value, it's best to change the battery settings in the following order:

Cut-off grid absent, Cut-off grid present, Re-discharge grid absent, Re-discharge grid present, Float, Bulk.

5.4 Meter correction

We can represent meter correction usage as below:

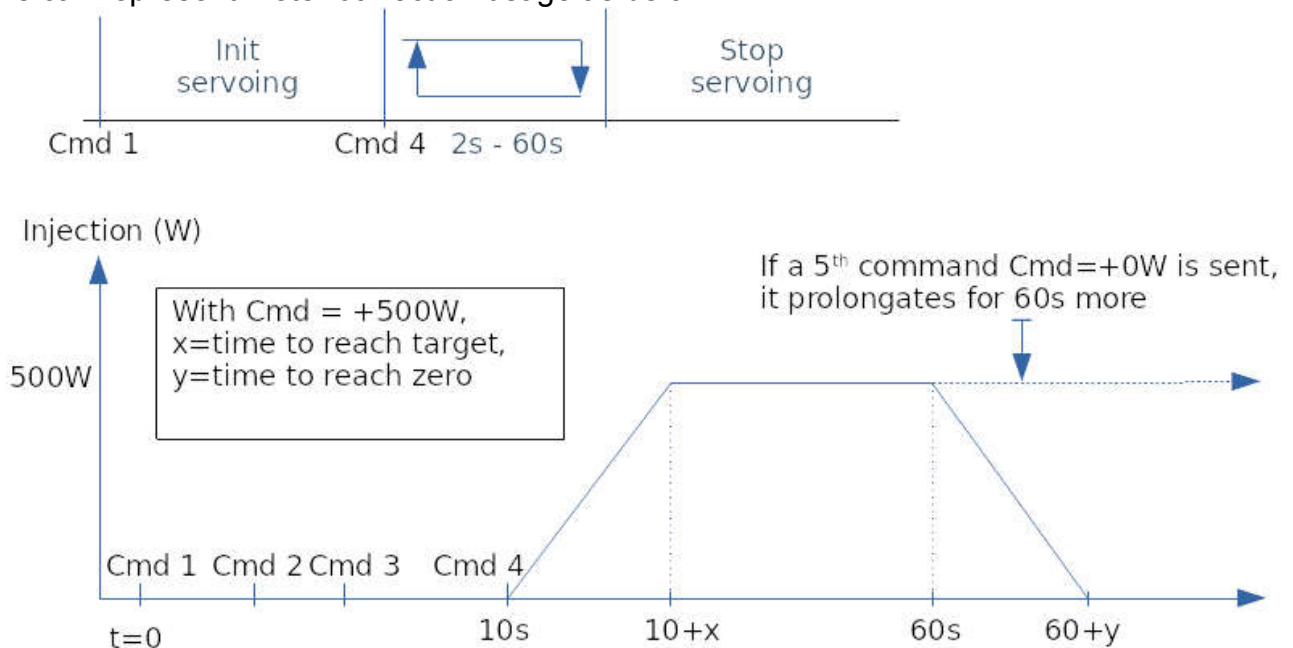


Figure 3. Meter correction behaviour

Meter correction requires an initialisation phase where meter correction commands (i.e. 0x140D, 0x1316, 0x1319, 0x131C, 0x131F, 0x1421) are sent with a regular period between 2 and 7 seconds over a period of 10s, so usually it means 3 or 4 writing commands. Once the initialisation is done, any command will be applied with respect to the grid allowed correction speed. We advise against using in the same time 0x1316 and the other meter commands, since their priority can lead to a conflict. The last command sent stills during 60s. If ever you need to shorten this time to stop injection, you can call 0x1100 to limit injection to 0 immediatly. After 60s, the inverter quits the meter correction and stops grid injection. In order to activate servoing mode, you need to send 0x140D multiple times during at least 10s.

5.5 Frequency and Voltage results of the Auto test for the CEI compliance

Here below, two tables with the description of the results of the Auto test for the frequency and the voltage for every phase of inverter's AC Grid input.

Results of Frequency test		
81<S2 Frequency Test		
Description	Type	Resolution
LimitFrequency	"short" encoded on 2 Bytes	0.1Hz
Limit Time	"short" encoded on 2 Bytes	1ms
Real Frequency	"short" encoded on 2 Bytes	0.1Hz
Trip Frequency	"short" encoded on 2 Bytes	0.1Hz
Trip Time	"short" encoded on 2 Bytes	1ms
81>S2 Frequency Test		
Description	Type	Resolution
LimitFrequency	"short" encoded on 2 Bytes	0.1Hz
Limit Time	"short" encoded on 2 Bytes	1ms
Real Frequency	"short" encoded on 2 Bytes	0.1Hz
Trip Frequency	"short" encoded on 2 Bytes	0.1Hz
Trip Time	"short" encoded on 2 Bytes	1ms

Results of Voltage test		
81<S2 Voltage Test		
Description	Type	Resolution
LimitFrequency	"short" encoded on 2 Bytes	0.1V
Limit Time	"short" encoded on 2 Bytes	1ms
Real Frequency	"short" encoded on 2 Bytes	0.1V
Trip Frequency	"short" encoded on 2 Bytes	0.1V
Trip Time	"short" encoded on 2 Bytes	1ms
81>S2 VoltageTest		
Description	Type	Resolution
LimitFrequency	"short" encoded on 2 Bytes	0.1V
Limit Time	"short" encoded on 2 Bytes	1ms
Real Frequency	"short" encoded on 2 Bytes	0.1V
Trip Frequency	"short" encoded on 2 Bytes	0.1V
Trip Time	"short" encoded on 2 Bytes	1ms

81<S1 Frequency Test		
Description	Type	Resolution
LimitFrequency	"short" encoded on 2 Bytes	0.1Hz
Limit Time	"short" encoded on 2 Bytes	1ms
Real Frequency	"short" encoded on 2 Bytes	0.1Hz
Trip Frequency	"short" encoded on 2 Bytes	0.1Hz
Trip Time	"short" encoded on 2 Bytes	1ms
81>S1 Frequency Test		
Description	Type	Resolution
LimitFrequency	"short" encoded on 2 Bytes	0.1Hz
Limit Time	"short" encoded on 2 Bytes	1ms
Real Frequency	"short" encoded on 2 Bytes	0.1Hz
Trip Frequency	"short" encoded on 2 Bytes	0.1Hz
Trip Time	"short" encoded on 2 Bytes	1ms

81<S1 VoltageTest		
Description	Type	Resolution
LimitFrequency	"short" encoded on 2 Bytes	0.1V
Limit Time	"short" encoded on 2 Bytes	1ms
Real Frequency	"short" encoded on 2 Bytes	0.1V
Trip Frequency	"short" encoded on 2 Bytes	0.1V
Trip Time	"short" encoded on 2 Bytes	1ms
81>S1 VoltageTest		
Description	Type	Resolution
LimitFrequency	"short" encoded on 2 Bytes	0.1V
Limit Time	"short" encoded on 2 Bytes	1ms
Real Frequency	"short" encoded on 2 Bytes	0.1V
Trip Frequency	"short" encoded on 2 Bytes	0.1V
Trip Time	"short" encoded on 2 Bytes	1ms

Table 15. Frequency and Voltage results of the Auto test

6. Error/Warning codes

Code	Type	Description
2	Warning	Islanding detected.
3	Warning	Grid voltage higher than the limit of the norm.
4	Warning	Grid voltage lower than the limit of the norm.
5	Warning	Grid frequency higher than the limit of the norm.
6	Warning	Grid frequency lower than the limit of the norm.
7	Warning	Grid voltage lost (Electrical grid loss)
8	Warning	Grid frequency lost (Electrical grid loss)
9	Warning	Grid three-phase dislocation.
10	Warning	Grid sinus lost (Electrical grid loss)
11	Warning	Average grid voltage greater than norm voltage for 10 min
12	Warning	Overload on AC Output with Grid.
13	Warning	Internal temperature nears the inverter temperature limit.
14	Warning	Earth fault.
15	Warning	Fan fault.
16	Warning	Software failure (Watchdog services: default B-Board service has crashed).
17	Warning	RAM usage over 70% of available capacity.
18	Warning	CPU usage over 70% of available capacity.
19	Warning	PV disconnected.
20	Warning	PV2 disconnected.
21	Warning	Low PV voltage.
22	Warning	Low PV 2 voltage.
23	Warning	PV voltage too high.
24	Warning	PV2 voltage too high.
25	Warning	CPU flash fail.
26	Warning	CPU initialization failure.
27	Warning	A battery warning has been reported.
28	Warning	Battery voltage under the cut-off voltage.
29	Warning	Battery voltage nears the cut-off voltage.
30	Warning	Battery not connected.
31	Warning	Battery discharge voltage low, near to the battery voltage detection threshold.
32	Warning	Battery voltage too high.
33	Warning	Battery voltage nears the cut-off voltage with grid.
34	Warning	Battery under the cut-off voltage with grid.
35	Error	CPU communication failure.
36	Error	CPU data communication failure.
37	Error	Internal wiring failure.
38	Error	Software critical failure (Watchdog services: Critical B-Board service has crashed).
39	Error	Coremod B-Board failure (watchdog).
40	Error	OS B-Board failure (watchdog).
41	Error	Failure reading/writing on database.
42	Error	Solar input overvoltage.
43	Error	Solar input overcurrent.
44	Error	Solar isolation resistance too low.
45	Error	DC Bus overvoltage.
46	Error	DC Bus undervoltage.

Code	Type	Description
47	Error	DC Bus circuit start timeout.
48	Error	Inverter voltage start timeout.
49	Error	Inverter overcurrent.
50	Error	Overheating.
51	Error	Internal relay failure.
52	Error	DC current sensor failure.
53	Error	PV input overvoltage.
54	Error	Over leakage current.
55	Error	Inverter DC overcurrent due to grid fluctuations.
56	Error	Grid measurement error on CPU.
57	Error	GFCI sensor failure.
58	Error	AC input ground loss.
61	Error	AC Output overload.
63	Error	AC Output short-circuit.
64	Error	Fan failure.
68	Error	AC Output undervoltage.
69	Error	AC Output overvoltage.
70	Error	Battery overvoltage.
71	Error	Battery disconnected.
74	Error	Auxiliary internal supply voltage failure.
87	Error	Battery current exceeds max battery current limit.
88	Error	Power on grid connection overpasses maximum power specification.
89	Error	Current on grid connection overpasses maximum current specification.
90	Error	Frequency on AC Output exceeds minimum frequency specification.
91	Error	Frequency on AC Output overpasses maximum frequency specification.
92	Error	Current on AC Output overpasses maximum current specification.
93	Error	PV input power overpasses specification.
94	Error	PV2 input power overpasses specification.
95	Error	Temperature exceeds the minimum specification of the inverter.
96	Error	Battery discharging current overpasses specification of the inverter.
97	Warning	Communication between Inverter and Smartmeter lost.
98	Error	Voltage on grid connection exceeds minimum voltage specification.
99	Error	Voltage on grid connection overpasses maximum voltage specification.
100	Error	Frequency on grid connection exceeds minimum frequency specification.
101	Error	Frequency on grid connection overpasses maximum frequency specification.
102	Error	Voltage on AC Output exceeds minimum voltage specification.
103	Error	Voltage on AC Output overpasses maximum voltage specification.
104	Error	Power on AC Output overpasses maximum power specification.
105	Error	Voltage on DC bus overpasses maximum voltage specification.
106	Error	Voltage on Battery overpasses maximum voltage specification.
107	Error	PV input current overpasses specification.
108	Error	PV2 input current overpasses specification.
109	Error	PV input voltage overpasses specification.
110	Error	PV2 input voltage overpasses specification.
111	Error	Battery charging current overpasses specification of the inverter.
112	Error	Neutral / Ground wires are reversed.
118	Error	Error happened during the Software package update.
119	Error	Temperature exceeds the maximum specification of the inverter.
136	Error	Transformer overheating.

Code	Type	Description
137	Error	DC Bus primary side overheating.
138	Error	PV boost board overheating.
139	Error	DC Bus secondary side overheating.
140	Error	Main board overheating.

Table 16 Fault/Warning codes

RESTRICTED

7. DRED mapping

The correspondance between IMEON ModBus protocol and Demande Response Enabling (DRED) functions are defined below.

Mode	Requirement	Function to implement
DRM 0	Operate the disconnection device.	1. Set AS4777 standard with register address 0x1322. 2. To disconnect the devise, open the relay with register address 0x1321.
DRM 1	Do not consume power.	1. Set DRM 0 (see step above) 2. Disable the AC input by opening relay with register address 0x1321.
DRM 2	Do not consume at more than 50% of rated power.	1. Set DRM0 (see step above) 2. When value from AC output is more than 50% (value available with register address 0x209), disable the AC input by opening relay with register address 0x1321.
DRM 3	Do not consume at more than 75% of rated power and source reactive power if capable	1. Set DRM0 (see step above) 2. When value from AC output is more than 75% (value available with register address 0x209), disable the AC input by opening relay with register address 0x1321.
DRM 4	Increase power consumption (subject to constraints from other active DRMs)	1. Set the current battery to limit the power to comply with other DRM with register address 0x1304 <i>Note : Power = current set (0x1304) x battery charging voltage (0x1307)</i> 2. Charge the battery by AC with register 0x1306 (bit 14) 3. Force continuous charge of the battery by setting time to charge battery to 00:00 to 00:00 with register address 0x1309
DRM 5	Do not generate power.	1. Disable the feed-in to the grid with register address 0x1306 (bit 13).
DRM 6	Do not generate at more than 50% of rated power.	1. Allow feed in to the grid with register address 0x1306 (bit 13). 2. Limit the feed in to the grid with register address 0x1100.
DRM 7	Do not generate at more than 75% of rated power and skin reactive power if capable.	1. Allow feed in to the grid with register address 0x1306 (bit 13). 2. Limit the feed in to the grid with register address 0x1100.
DRM 8	Increase power generation (subject to constraints from other active DRMs)	1. Allow feed in to the grid with register address 0x1306 (bit 13). 2. Allow the battery to discharge to the grid with register address 0x1306 (bit 9 and 10) 3. Limit the feed in to the grid with register address 0x1100 to comply with other DRM.

RESTRICTED

8. Frame examples

8.1 Identification frame

The function code to use is 0x9000. This function code is used only once at start of communication with inverter. The value to send is a key of eight bytes given by IMEON ENERGY.

Device: [01 10 90 00 00 04 08 XX XXXXXXXXXXXXXXXX CRCL CRH]

IMEON: [01 10 90 00 00 04 80 CRCL CRCH]

8.2 Get the serial number of inverter

The function code to use is 0x0001:

Device: [01 03 00 01 00 04 15 C9]

IMEON: [01 03 00 08 00 00 00 00 07 5B CD 15 14 BD]

Result: HEX [0x00 00 00 00 07 5B CD 15] to DEC [123456789]. Serial number: 123456789.

8.3 Get the date of inverter

The function code to use is 0x1001:

Device: [01 03 10 01 00 04 11 09]

IMEON: [01 03 08 00 E1 07 0A 02 11 02 0E 2C 76]

Result: HEX [0x00 E1 07 0A 02 11 02 0E] to DEC [2017][10][02][15][02][14]. Date is: 2017/10/02 15:02:14.